

BioHarvest-Sim: Analytical Assessment Report

Species: Fin Whale | Generated: 2026-09-07 12:08

ISO/IEC/IEEE 12207 Compliance | High-Precision Linear Algebra Core

EXECUTIVE SUMMARY:

This report details the demographic stability and sustainable yield optimization for 'Fin Whale' based on discrete-time Leslie matrix theory. The population exhibits a dominant eigenvalue $\lambda_1 = 1.0492$, corresponding to an intrinsic growth rate of $r = 0.0481$ per year. The net reproductive rate R_0 is 1.5072 with a mean generation time of 8.54 years. The theoretical maximum uniform sustainable harvest fraction is 4.69% per annum.

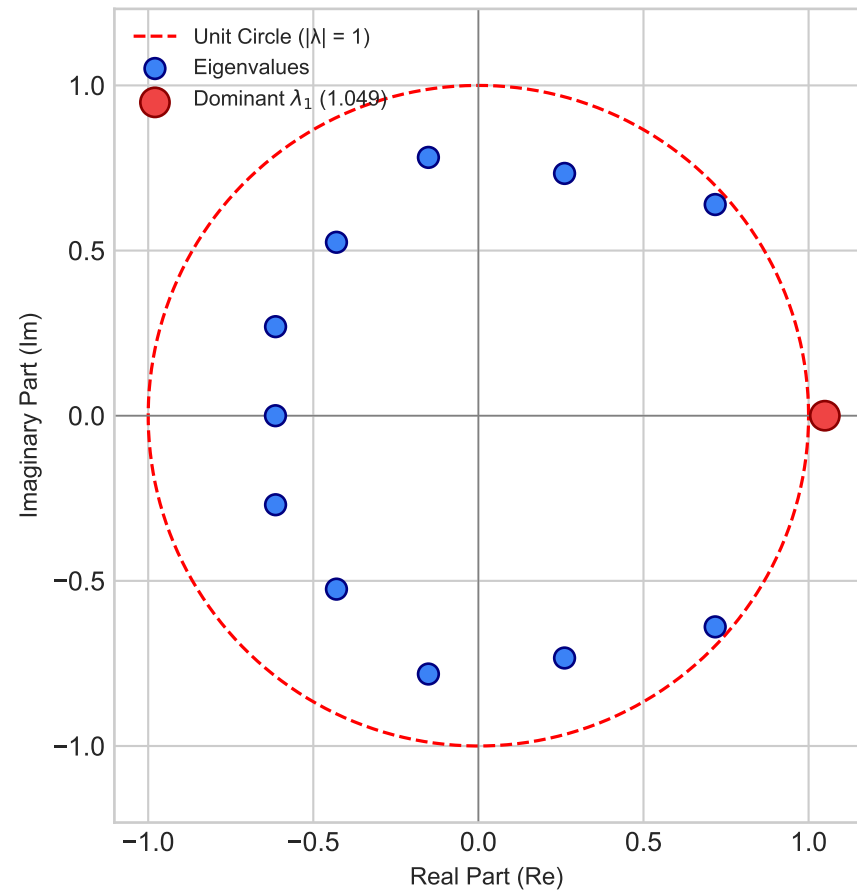
Age Classes (N)	12
Dominant Eigenvalue (λ_1)	1.04924
Intrinsic Growth Rate ($r = \ln \lambda_1$)	0.04806 yr ⁻¹
Net Reproductive Rate (R_0)	1.50715
Mean Generation Time (Tc)	8.54 years
Damping Ratio ($\lambda_1 / \lambda_2 $)	1.0921
Perron-Frobenius Primitivity	Verified Primitive
Matrix Diagonalizable	Yes (L = P D P ⁻¹)
Uniform Sustainable Harvest (h*)	4.69% per step

Leslie Projection Matrix (L):

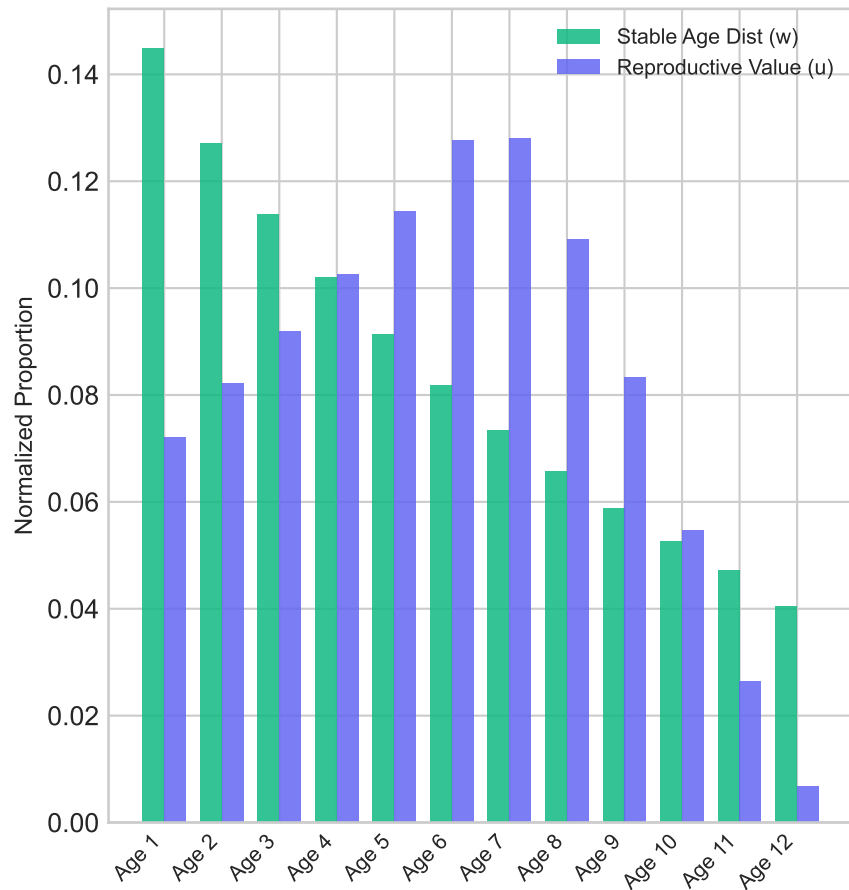
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[ [0.    0.    0.    0.    0.    0.19 0.44 0.5  0.5  0.45 0.3  0.1 ]
  [0.92 0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0. ]
  [0.    0.94 0.    0.    0.    0.    0.    0.    0.    0.    0.    0. ]
  [0.    0.    0.94 0.    0.    0.    0.    0.    0.    0.    0.    0. ]
  [0.    0.    0.    0.94 0.    0.    0.    0.    0.    0.    0.    0. ]
  [0.    0.    0.    0.    0.94 0.    0.    0.    0.    0.    0.    0. ]
  [0.    0.    0.    0.    0.    0.94 0.    0.    0.    0.    0.    0. ]
  [0.    0.    0.    0.    0.    0.    0.94 0.    0.    0.    0.    0. ]
  [0.    0.    0.    0.    0.    0.    0.    0.94 0.    0.    0.    0. ]
  [0.    0.    0.    0.    0.    0.    0.    0.    0.94 0.    0.    0. ]
  [0.    0.    0.    0.    0.    0.    0.    0.    0.    0.94 0.    0. ]
  [0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.9 0. ] ]
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Spectral Decomposition & Asymptotic Age Structure

Eigen-Spectrum (Complex Plane)

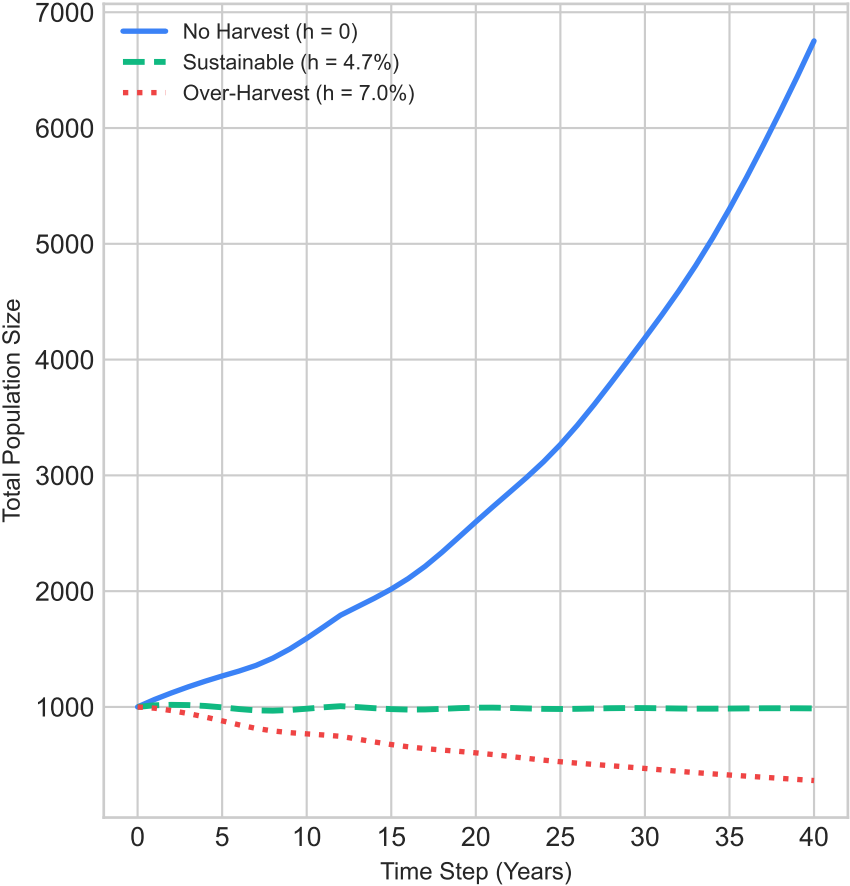


Stable Distribution vs. Reproductive Value

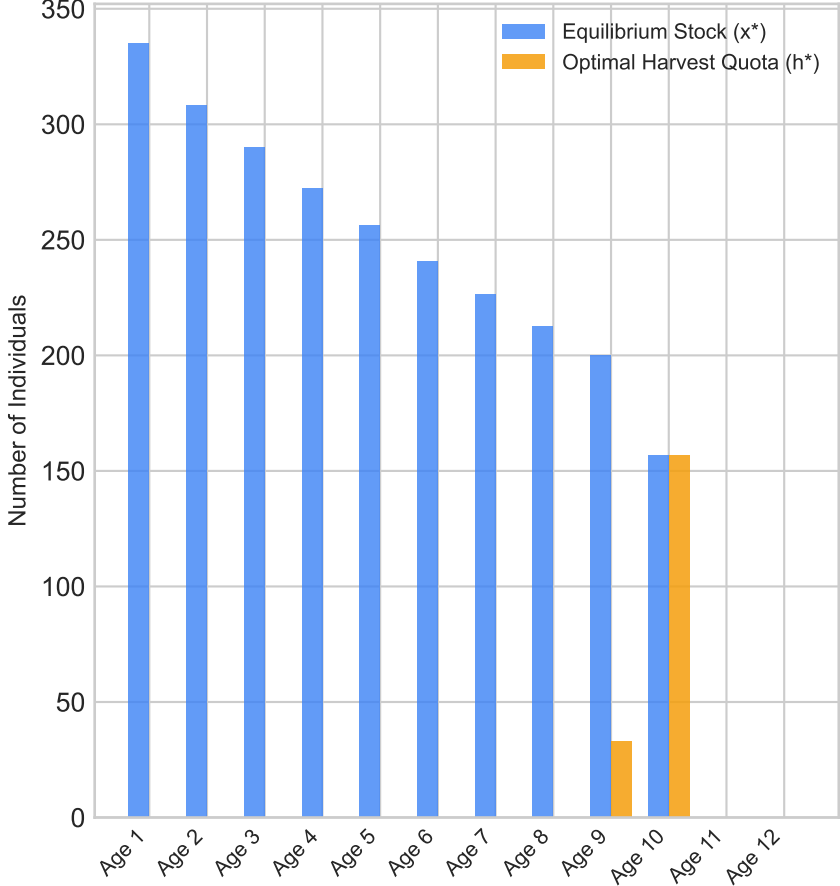


Harvest Dynamics & Optimization

Population Trajectory Comparison

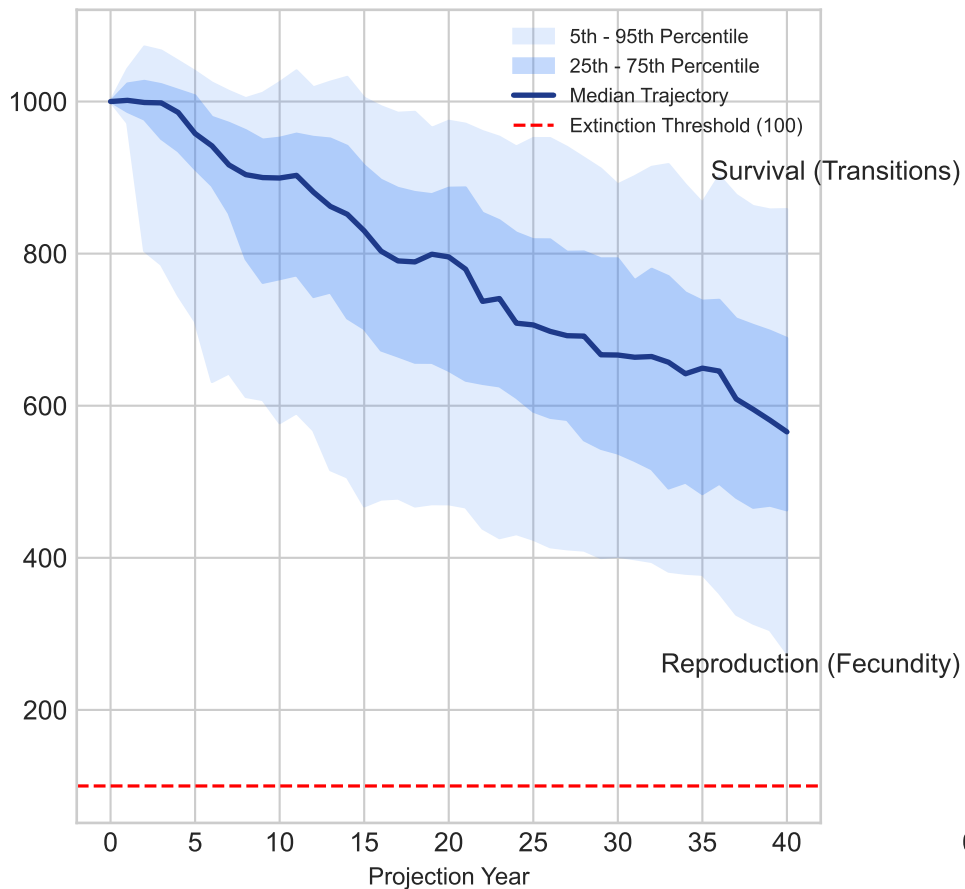


MSY Linear Program Solved (Yield: 190.0)



Environmental Stochasticity & Vital Rate Elasticities

Monte Carlo Extinction Risk (0.0%)



Elasticity Contribution to Growth (λ_1)

